

# Enterprise VLAN & Trunking Lab

## Phase 1 Report – Network Planning and Topology Design

### Project Information

**Project Name:** Enterprise VLAN & Trunking Lab

**Phase:** Phase 1 – Network Planning and Topology Design

**Simulation Software:** Cisco Packet Tracer

**Project Type:** Enterprise LAN Design

**Status:** Completed

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## 1. Introduction

The first phase of the Enterprise VLAN & Trunking Lab focused on designing and preparing the network infrastructure before any switch configuration was performed. Proper planning is a fundamental step in network engineering because it minimizes configuration errors, simplifies troubleshooting, and provides a clear implementation roadmap.

During this phase, the enterprise network topology was created, network devices were placed and interconnected, and an IP addressing plan was established. No VLANs or switch configurations were applied at this stage.

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## 2. Objectives

The objectives of this phase were to:

- Design a scalable enterprise LAN topology.
  - Simulate a multi-department corporate network.
  - Place and organize all network devices.
  - Connect devices using the appropriate cable types.
  - Develop an IP addressing scheme.
  - Prepare the environment for VLAN implementation.
  - Save the initial network design.
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### 3. Business Scenario

TechCorp is expanding its internal network infrastructure to support four independent departments:

- Management
- Human Resources (HR)
- Sales
- Information Technology (IT)

To improve network security, reduce unnecessary broadcast traffic, and simplify administration, each department will eventually operate within its own VLAN. This phase establishes the physical network required to support that logical segmentation.

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### 4. Network Devices

The following devices were deployed in the simulation:

| Device Type        | Quantity |
|--------------------|----------|
| Cisco 2960 Switch  | 2        |
| Personal Computers | 8        |

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### 5. Device Naming

#### Switches

| Device  | Hostname |
|---------|----------|
| Switch0 | SW1      |
| Switch1 | SW2      |

#### End Devices

| Device | Hostname |
|--------|----------|
| PC0    | MGMT-PC1 |
| PC1    | MGMT-PC2 |
| PC2    | HR-PC1   |
| PC3    | HR-PC2   |

| Device | Hostname  |
|--------|-----------|
| PC4    | SALES-PC1 |
| PC5    | SALES-PC2 |
| PC6    | IT-PC1    |
| PC7    | IT-PC2    |

Using descriptive hostnames improves readability, troubleshooting efficiency, and configuration management.

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## 6. Physical Topology

The enterprise network consists of two Cisco switches connected through a Gigabit Ethernet uplink.

Each switch serves a different group of end devices.

### Switch Connections

- SW1 GigabitEthernet0/1 ↔ SW2 GigabitEthernet0/1

### End Device Connections

#### SW1

| Port  | Connected Device |
|-------|------------------|
| Fa0/1 | MGMT-PC1         |
| Fa0/2 | MGMT-PC2         |
| Fa0/3 | HR-PC1           |
| Fa0/4 | HR-PC2           |

#### SW2

| Port  | Connected Device |
|-------|------------------|
| Fa0/1 | SALES-PC1        |
| Fa0/2 | SALES-PC2        |
| Fa0/3 | IT-PC1           |
| Fa0/4 | IT-PC2           |

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## 7. IP Addressing Plan

| Department | Device    | IP Address    | Subnet Mask   |
|------------|-----------|---------------|---------------|
| Management | MGMT-PC1  | 192.168.10.10 | 255.255.255.0 |
| Management | MGMT-PC2  | 192.168.10.20 | 255.255.255.0 |
| HR         | HR-PC1    | 192.168.20.10 | 255.255.255.0 |
| HR         | HR-PC2    | 192.168.20.20 | 255.255.255.0 |
| Sales      | SALES-PC1 | 192.168.30.10 | 255.255.255.0 |
| Sales      | SALES-PC2 | 192.168.30.20 | 255.255.255.0 |
| IT         | IT-PC1    | 192.168.40.10 | 255.255.255.0 |
| IT         | IT-PC2    | 192.168.40.20 | 255.255.255.0 |

No default gateway was configured because inter-VLAN routing is not implemented in this phase.

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## 8. Design Considerations

The network was designed with the following principles:

- Modular and scalable topology.
  - Separation of departments through future VLAN implementation.
  - Consistent naming convention.
  - Structured IP addressing.
  - Dedicated uplink between switches for VLAN trunking.
  - Readiness for future implementation of routing, security, and network services.
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## 9. Deliverables

The following deliverables were completed:

- Enterprise topology created in Cisco Packet Tracer.
- All devices placed and renamed.
- Physical cabling completed.
- Inter-switch uplink established.
- Static IP addresses assigned to all end devices.
- Initial Packet Tracer project saved.

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## 10. Verification

The following checks were performed:

- All devices are present in the topology.
- PCs are connected to the intended switch ports.
- Switches are interconnected through GigabitEthernet0/1.
- IP addresses match the addressing plan.
- Packet Tracer project opens without errors.

At this stage, communication between devices is not expected because VLANs have not yet been created and switch ports remain in the default VLAN.

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## 11. Skills Acquired

By completing this phase, the following skills were practiced:

- Enterprise network planning
  - Physical topology design
  - Device naming conventions
  - Structured cabling
  - IP addressing and subnetting
  - Documentation of network infrastructure
  - Preparation for Layer 2 switch configuration
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## 12. Conclusion

Phase 1 successfully established the physical infrastructure required for the Enterprise VLAN & Trunking Lab. The topology, addressing plan, and device organization provide a solid foundation for implementing VLANs and IEEE 802.1Q trunking in the next phase. This planning-first approach reflects industry best practices and ensures that subsequent configuration tasks can be performed in a structured and maintainable manner.